

MST-Technologies and Processes

September, 29th, 2020, 08:00
G.-Köhler-Allee 101, HS 00-026, SR 00-010/14

Duration: 120 min.

Please note

- write your name and matriculation number into the caption of every sheet
- use and write only on the front of the sheet
- use a separate sheet for every task
- write with a pen NOT with a pencil

Cheating

- any attempt of cheating results in immediate failure of the exam
- switch off your cell phones and store them in your backpack – a cell phone on the table will be treated as an attempt to cheating

Withdrawal or absence

- you may withdraw until the moment the tasks are revealed
- please see:
https://www.tf.uni-freiburg.de/en/studies-and-teaching/a-to-z-study-faq/de-registration-of-exams?set_language=en
- you find the relevant forms here:
<https://www.tf.uni-freiburg.de/de/studium-lehre/a-bis-z-studium/dokumente/WithdrawalfromExamsduetoIllnessMasterofScience.pdf>

Necessary Points: 50 of 115 in total

Points: _____

Result: _____

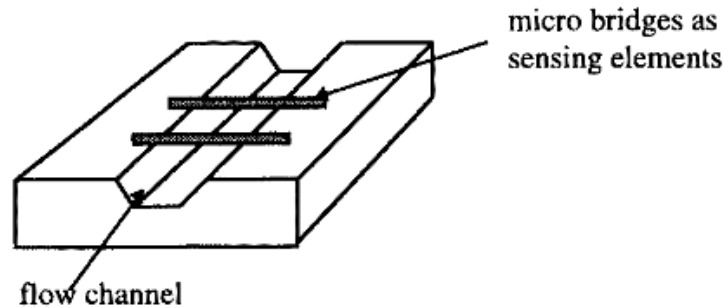


1. Processes

20

Sketch the manufacturing steps required to fabricate the following structure (an anemometric flow sensor). The substrate is a (100) silicon wafer, the micro bridges are made from gold.

Sketch and briefly explain each step.



2. Cleanroom technology

26

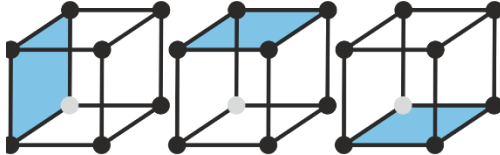
- | | | |
|-----|--|----|
| 2.1 | Name four general recommendations for working in the cleanroom | 4 |
| 2.2 | What are common ISO classes used for white areas in a cleanroom for MEMS and for microelectronics? | 4 |
| 2.3 | What are the two common cleanroom setups? Sketch a typical setup for each of the two and give one advantage and one disadvantage for each of the two | 10 |
| 2.4 | What are the four filtering concepts used in filter-fan units? Explain briefly how each of them functions | 8 |



3. Semiconductor materials

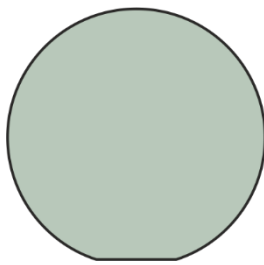
22

- 3.1 Name two common semiconductor materials 2
- 3.2 What is their most outstanding physical properties (for which they would be most likely chosen in an application) 2
- 3.3 Name and sketch 4 different Bravais lattices 8
- 3.4 Indicate the following crystal planes by their Miller indices 6

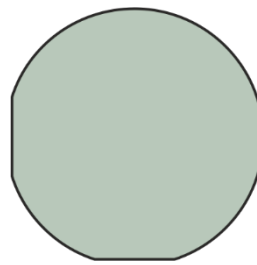


- 3.5 State the wafer type for the given flats (assume p doping on both cases) 4

a)



b)



4. Photolithography

27

- 4.1 Sketch and explain the process workflow for generating a chromium mask using a mask-less lithography 10
- 4.2 Why are anti-reflective coatings used in photolithography? 2
- 4.3 Which thickness is required for an anti-reflective coating made from a material with $n=1.5$ for a wavelength of 600 nm 4
- 4.4 Name three common substrate materials for photomasks 3
- 4.5 Sketch the lithographic setup in proximity lithography 6
- 4.6 Name the two relevant classes of photoresists 2



5. Etching

20

- 5.1 Sketch and explain two workflows for generating a metallic electrode structure (e.g., made from gold); one of these processes will require a metal etch, the other does not 10
- 5.2 State the wet etching reaction of silicon in a mixture of nitric acid and hydrofluoric acid (HF) 4
- 5.3 Given the following etching diagram state three etching solutions which give an etching rate of 13 ($\mu\text{m}/\text{min}$) in acetic acid (solid lines) as diluent 6

